Census structure of floor-attaining local models for the singlet

Supporting computations for arXiv:2608.18886 — deposit note, version 1

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Question

In the moiré-phase-locking mechanism of arXiv:2608.18886 (Sec. 7), each event carries frozen per-station offsets $\left(\xi_A, \xi_B\right)$, a flight time τ and a coupling scale, drawn from a setting-independent density ρ. On 1024 random or stratified offsets at the 161-angle register and tolerance $\varepsilon = 0.01$ the linear program returns $M/M_{\min} = 1.000$, with $M_{\min} = \left(\sqrt{2} - 1\right)/3$ Hall's floor. The paper states that ρ is optimizer-chosen and that deriving it from a formation process is open. This deposit asks what the floor-attaining ρ looks like, whether it depends on the terrain U, and whether it can be built from the two stations separately.

Method

Four independent floor-attaining solutions were produced with `dither_sweep_rho.py` (NANG = 161, NSH = 32, i.e. 1024 offsets, $\varepsilon = 0.01$, SAVE_RHO = 1): random seeds 0 and 1, stratified seeds 0 and 1. Each has $M/M_{\min} = 1.000$ to twelve digits; the four share no offsets. `census_structure_v2.py` forms the weighted empirical measure over $\left(\xi_A, \xi_B\right)$, pooled over τ and scale, and its circular Fourier plane $\left|F\left(k_a, k_b\right)\right|$ for $|k| \le 6$, with phases; the null is a 400-draw seeded permutation (weights fixed, offsets re-paired). `terrain_factor_v1.py` tests whether the 23-term terrain factors as two screens. `flight_time_perturb_v1.py` / `dither_sweep_perturb_v1.py` re-solve on a perturbed terrain with the same 1024 offsets as random0. `product_census_v3.py` tests, on the random0 banks, whether a census of the form $q\left(\tau, s\right) f\left(\xi_A\right) g\left(\xi_B\right)$ ("product"), or one with ξ_B independent of everything else ("semi"), reproduces the singlet correlators at tolerance ε, against a binned joint census as control; `breed_product_v1.py` with `pipeline_v1.py` breeds a terrain scored by the product-class price and validates the champion at full scale.

Result A — a universal low-order skeleton (four censuses)

census	active atoms	anti-diag $\left(k, -k\right)$ z	diag $\left(k, k\right)$ z	top pixel $\lvert F\rvert$	2nd pixel $\lvert F\rvert$
random0	658	+0.9	+5.8	(2,2) 0.373	(2,0) 0.244
random1	672	−0.7	+6.2	(2,2) 0.352	(2,0) 0.293
strat0	—	−0.7	+5.8	(2,2) 0.349	(2,0) 0.310
strat1	—	−0.9	+7.1	(2,2) 0.384	(2,0) 0.251

Null 95th percentile for the top pixel: 0.17–0.18 on every run. Phases φ of $F\left(k_a, k_b\right)$ (the census contains $|F| \cos\left(2k_a\xi_A + 2k_b\xi_B - \varphi\right)$):

pixel	random0	random1	strat0	strat1	spread	P(4 uniform phases in that arc)
(2,2)	+156.5°	+161.1°	+157.1°	+158.0°	4.5°	8×10^{-6}
(2,0)	+148.2°	+148.9°	+154.5°	+152.3°	6.3°	2×10^{-5}

pixel	random0	random1	strat0	strat1	spread	P(4 uniform phases in that arc)
(4,−2)	−157.9°	−155.1°	−160.1°	−159.1°	4.9°	1×10^{-5}
(2,4)	−84.9°	−82.6°	−68.8°	−83.6°	16°	4×10^{-4}

Statements supported, in descending strength: (1) ρ^* is not a function of $\xi_A - \xi_B$ alone (anti-diagonal share at the null, four times); a U(1)-symmetric common-cause census with only relative-phase correlation does not attain the floor on this terrain. (2) ρ^* carries a sum term $\cos 4\left(\xi_A + \xi_B - \varphi\right)$, $\varphi \approx 158°$, as its dominant structure and a nonuniform ξ_A marginal $\cos 4\xi_A$; magnitudes and phases agree across two samplers and two seeds sharing no atoms. (3) The A/B asymmetry ((2,0) > (0,2)) is a property of the solve, which pins $a = 0$ and sweeps b, not of the stations. (4) Weight sits at short flights (random0: $\tau = 0.5$ carries 37%, $\tau = 1$ 27%).

Result B — the terrain is not a product of two screens

On `greedy_results.json` (23 terms, $v = 15.2339$): a product of two real cosine screens predicts $w_{mn+} = w_{mn-}$ for every (m, n); observed ratios run 0.05 to 183. A rank-1 complex-screen fit within each sign class misses by 2.3–2.8× rms. Difference beats ($k = m - n$) carry 72% of the weight. This is a statement about the greedy-found terrain only.

Result C — terrain perturbation (matched control: random0, same 1024 offsets)

run	terrain change	$M - M_{\min}$	(2,2) $\lvert F\rvert$, φ	(2,0)	(4,−2)	new in top 5
random0 (control)	none	10^{-13}	0.373, +156.5°	0.244, +148.2°	0.184, −157.9°	—
A	(2,2,+) phase +45°	5×10^{-8}	0.422, +118.6°	0.258, +127.6°	0.199, −169.7°	(2,−2) 0.237, (0,4) 0.189
B	(2,2,+) removed	3×10^{-14}	0.307, −144.5°	—	0.206, −165.6°	(2,4) 0.285, (0,2) 0.253

The floor does not depend on the (2,2,+) term; the census does. Under a +45° shift of the term's phase the (2,2) pixel moves by −38° (the sign the convention $\sin(2k\lambda - \phi)$ dictates). With the term removed a (2,2) pixel persists at 0.31 at a different phase, from harmonic mixing of the remaining terms, so the pixel tracks the term but is not solely owned by it. The census skeleton is a function of the terrain; the price is not.

Result D — station-local censuses do not reach the floor

On the random0 banks (`product_census_v3.py`, NB = 16 and 32 bins, 20 starts per class, $\varepsilon \in \{0.10, 0.05, 0.02, 0.01\}$): no product census and no semi census reproduces the correlators at $\varepsilon \leq 0.02$ (final correlator violations 2–15%), while the binned joint census does (1.59× the floor at NB = 16 and 1.12× at NB = 32 at $\varepsilon = 0.01$, the excess being the binning). Where the station-local classes are feasible, $\varepsilon \geq 0.05$, they pay 2.2–3.0× the joint price. Fourier checks on the best product censuses confirm they are honest products ($\lvert F(2,2)\rvert \approx \lvert F(2,0)\rvert \lvert F(0,2)\rvert$).

Result E — a terrain bred to favour station-local censuses does not change this

`breed_product_v1.py` from the greedy terrain (breeding scale NSH = 8, NANG = 9, $\varepsilon = 0.10$) stopped after five rounds with no improving move: joint 1.631×, product 1.927× the floor, ratio 1.18 — "promising" under the pre-

registered rule (feasible and ratio ≤ 1.3). Full-scale validation of the champion (NANG = 161, 1024 random offsets, NB = 32, 12 starts): at $\varepsilon = 0.05$ joint 1.261, semi 2.492, product 2.864; at $\varepsilon = 0.01$ joint 1.591, semi and product infeasible. The breed improved the product class only relative to the joint class at coarse scale, by degrading the joint; it never brought a product census near the floor and the gain vanished at full resolution. Stage B (minimal start) was not run; the exclusion reads “from the greedy start”.

Standing conclusion

On the paper’s terrain, on four independent censuses, and on a terrain bred to favour station-local censuses, no product census reproduces the singlet at $\varepsilon \leq 0.02$, and where feasible it pays 2–3× the joint price. The frozen offsets of the two stations must be correlated at the source. This does not touch the disc–lattice interaction as a reading of U or of D; it excludes it as the origin of ρ. Nothing here derives ρ.

Caveats

The exclusion is computational and “on these terrains”, not a theorem; the theorem-level statement is Bell’s (station-side processing is $A\,(\,a, \lambda\,)$). The (2,2) pixel tracks the (2,2,+) term but is not solely owned by it. Results A (random0), B and the perturbation module’s identity check were reproduced on a second machine; the remaining runs are from the author’s workstation with the scripts’ internal self-checks (stored-ρ price reproduced to twelve digits before each run). `product_census_v1`/`v2` are withdrawn (v1 had a normalization error) and are not part of this deposit.

Not included in this deposit

The sum/difference-coordinate census test (`product_census_v4.py`, COORDS = sumdiff), running at the time of deposit; the minimal-start terrain breed; the Hamiltonian residence-law, profile-spectrum, jitter-damping and asymmetric-frame questions. These will appear in a later version of this record if run.

Files

`census_structure_v1.py`, `census_structure_v2.py`, `terrain_factor_v1.py`, `flight_time_perturb_v1.py`, `dither_sweep_perturb_v1.py`, `product_census_v3.py`, `product_census_v4.py`, `breed_product_v1.py`, `pipeline_v1.py`, `RUN_PIPELINE.bat`, `check_folders_v1.py`; `greedy_results.json`; `rho_nsh32_{random0,random1,strat0,strat1}_eps0.01_nang161.json`; `rho_perturb_2_2_1_{shift_45,weight_0}_nsh32_random0_eps0.01_nang161.json`; `banks_nsh32_random0_nang161_nw2048.npz`; `product_census_v3_random0_NB{16,32}_*.{json,txt}`; `breedrun/A_breed_greedy/{breed_product_results.json, bred_product_champion.json}`; `breedrun/VA_validate_greedy/`; `pipeline_FINAL.txt`; run transcripts. Every number quoted above is produced by the named script from the named input; the scripts assert the stored price before reporting.